IEC 60204-1 EMERGENCY STOP & HARDWARE INTERLOCK HIERARCHY

Deterministic 4-Tier Fallback Escalation from Real-Time 1 kHz CBF Filtering to Hardware Safe Torque Off (STO)

LEVEL 0: NOMINAL CBF

State: Closed-Loop Control

1000 Hz Minimal-Intervention QP

Activation Trigger:

- Safe set invariant satisfied: $h(x) \geq 0$
- Stopping margin clear: $d_{\text{clear}} > d_{\text{stop}}(v)$
- Valid unexpired intent lease: TTL > 0

Hardware & Actuator Consequence:

- Permitted setpoint: $u_t = \Pi_{\mathcal{U}_{\text{safe}}}(p_t)$
- 3-phase inverter gate drivers actively PWM modulated
- Full task execution authority

LEVEL 1: CATEGORY 2 STOP

Standard: IEC 60204-1 Cat 2

Controlled Position / Coordinate Hold

Activation Trigger:

- MPU Intent lease expired: $t > t_{\text{expire}}$
- IPC heartbeat missing: 15 ms $< \Delta t \leq$ 50 ms
- Non-critical sensor degraded / occluded

Hardware & Actuator Consequence:

- Actuator coils remain energized
- Closed-loop PID holds coordinate $x(t) = x_{\text{hold}}$
- System awaits fresh valid MPU intent lease

LEVEL 2: CATEGORY 1 STOP

Standard: IEC 60204-1 Cat 1 (SS1)

Safe Stop 1: Controlled Deceleration

Activation Trigger:

- Stopping clearance breach: $d_{\text{clear}} \leq d_{\text{stop}}(v)$
- Safe set boundary violated: $h(x) < 0$
- Joint kinematic / velocity envelope exceeded

Hardware & Actuator Consequence:

- Active maximum dynamic braking: $a_{\text{brake}} = a_{\max}$
- Mechanism brought to zero velocity ($v \to 0$)
- Actuator power removed after standstill is reached

LEVEL 3: CATEGORY 0 STOP

Standard: IEC 60204-1 Cat 0 (STO)

Safe Torque Off / Direct Power Cut

Activation Trigger:

- Physical E-Stop button pressed
- Hardware watchdog timer trip: $\Delta t >$ 50 ms
- Hardware over-current trip: $I_{\text{phase}} > I_{\text{trip}}$

Hardware & Actuator Consequence:

- Hardware optocoupler pulls `EN_GATE` low in $< 5\,\mu$s
- Zero software dependency; gate drive power isolated
- Spring-actuated mechanical friction brakes engage

↑ INCREASING SEVERITY & HARDWARE ISOLATION

The Asynchronous State Re-Anchoring Handshake Upon Safety Intervention

1. MCU Intercepts & Latches:

- Safety enforcer vetoes trajectory p_t
- MCU asserts atomic flag `IS_INTERVENTION = 1`
- Latches resting coordinate x_{veto} in SRAM

2. IPC Mailbox Notification:

- Hardware interrupt posted to Linux MPU
- Telemetry stream tagged with intervention ID
- Prevents DAgger covariate policy corruption

3. MPU Pipeline Flush & Reset:

- MPU aborts obsolete action chunk buffer
- Flushes transformer KV-cache
- Re-seeds diffusion planner from true x_{veto}